(1)

Problem 1: 4,35
Problem 2: 2,5
Problem 3: 1,175
Problem 4: 0,0
6,6

Thứ ngày tháng năm

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4,35 **Problem 1: Singly Linked List – In-Place Reversal and Pointer Tracing**

Given a singly linked list: head → [10] → [25] → [40] → [15] → NULL

Task 1: Write a function / algorithm in pseudocode to reverse the singly linked list, including three pointers (prev, curr, next) to complete your function.

Idea:

1,1 Initially: prev = NULL
curr = head

For each iteration, perform 4 steps:
1/ Save the next node in "next"
2/ Reverse the direction of "curr.next" so that it points to "prev"
3/ Move "prev" to "curr"
4/ Move "curr" to "next"

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Pseudocode:

```
Function ReverseList(Head)
    prev ← NULL
    curr ← head
While (curr ≠ NULL)
    next ← curr.next
    curr.next ← prev
    prev ← curr
    curr ← next
End While
    head ← prev
Return head
End Function
```

1.25 Task 2: Draw a diagram or step-by-step pointer trace table showing the positions and values of the three pointers after each iteration until the list is fully reversed

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Iteration	prev	curr	next
Initial	NULL	10	—
1	10	25	25
2	25	40	40
3	40	15	15
4	15	NULL	NULL

List after interation:

Initial: 10 -> 25 -> 40 -> 15 -> NULL

1 : 10 -> NULL

2 : 25 -> 10 -> NULL

3 : 40 -> 25 -> 10 -> NULL

4 : 15 -> 40 -> 25 -> 10 -> NULL

Final result:

head

1

[15] -> [40] -> [25] -> [10] -> NULL

2.5 Problem 2: Circular Linked List – Insert at Head

Given a singly circular linked list:

head -> [10] -> [20] -> [30] -> (circles back to head)

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Thứ ngày tháng năm

1,25

Task 1: Pseudocode

```
Function InsertHead (head, newValue)
    newNode ← CreateNode (newValue)
    If head = NULL then
        newNode.next ← newNode
        head ← newNode
        Return head
    End If
    tail ← head
    While (tail.next ≠ head) do
        tail ← tail.next
    End While
    newNode.next ← head
    tail.next ← newNode
    head ← newNode
    Return head
End Function
```


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(2)

Ngày .....tháng .....năm.....

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Problem 2:

1,25 Task 1: Trace when newValue = 5

Step 1: Initial state

head

[10] -> [20] -> [30]

Therefore: head = 10

Step 2: Create newNode

newNode = 5

Step 3: Find the tail

Initially: tail = head = 10

Check: tail.next != head

Since: 10.next = 20

Le move: tail = 20

Again: 20.next = 30

So: tail = 30

Now: 30.next = head

=> 30 is the tail

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Step 4: Connect the newNode to the old head

Execute:

newNode.next = head

Therefore: 5.next = 10

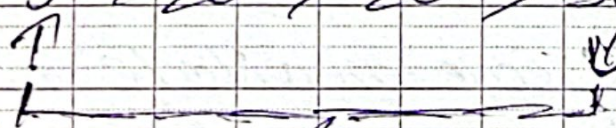
Now: [5] → [10] → [20] → [30]

Step 5: Connect the tail to the newNode

Execute: tail.next = newNode

Therefore: 30.next = 5

The circular structure becomes:

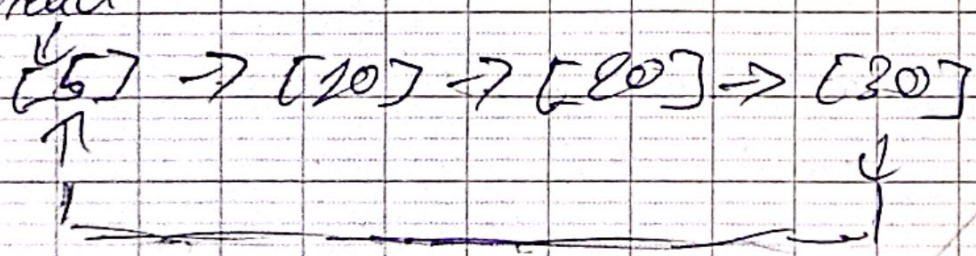


Step 6: Update head

head = newNode

Therefore: head = 5

Final result:



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Ngày .....tháng .....năm.....

1.75 Problem 3: Doubly linked list

1.75 Task 1: Pseudocode

Function: InsertBefore(head, target, newValue)
curr ← head
While (curr ≠ NULL and curr.data ≠ target)
curr ← curr.next
End While
If curr = NULL then
Return head
End if
newNode ← CreateNode(newValue)
If curr = head then
newNode.prev ← NULL
newNode.next ← head
head.prev ← newNode
head ← newNode

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Ngày .....tháng .....năm.....

```
ELSE
    newNode.prev ← curr.prev
    newNode.next ← curr
    curr.prev.next ← newNode
    curr.prev ← newNode
End If
Return head
End Function
```


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